Geostatistical Refinement of GIS Classification Results

Evaluating and Correcting Spatial Outliers via Kriging-based Hypothesis Testing

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Presentation Outline

1. Introduction & Background
2. Exploratory Data Analysis (EDA)
3. Methodology (Step-by-Step)
4. Experimental Results (Detailed)
5. Conclusion

Background: Pixel-based Classification in GIS

Tobler's 1st Law

"Everything is related to everything else, but near things are more related than distant things." [4]

Context: Remote Sensing (RS)

- High-resolution data demands automated, pixel-wise Land Use/Land Cover (LULC) classification.

The Flaw in EM: EM Clustering treat every single pixel as an independent sample. this directly violates Tobler's First Law of Geography. Natural landscapes are continuous and spatially autocorrelated.

Problem Statement: The 'Salt-and-Pepper' Effect

The Phenomenon: Spatially isolated pixels or fragmented clusters whose labels differ significantly from their immediate neighbors [1].

Root Causes: Intra-class heterogeneity (e.g., shadows, moisture variations) and sensor noise.

Consequences in GIS

- Degraded LULC decision-making.
- Inaccurate spatial area estimation.
- Hindered landscape ecology metrics.

Limitations of Heuristic Filters

- **Majority Filter:** Purely geometric; irreparably *destroys thin linear features* (e.g., roads, rivers) [3].
- **MRF:** Tends to oversmooth genuine geographic boundaries.
- **OBIA:** Relies heavily on subjective parameters.

Research Objectives & Pipeline

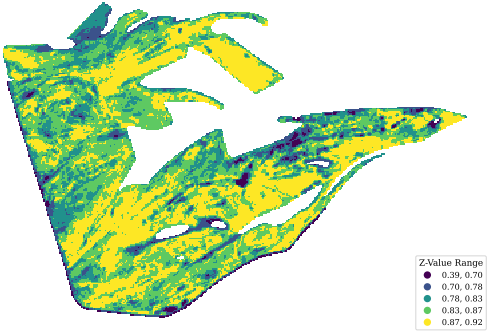
The Core Question

Is an assigned isolated class label **statistically consistent** with the underlying spatial distribution of the continuous physical attribute (Z)?

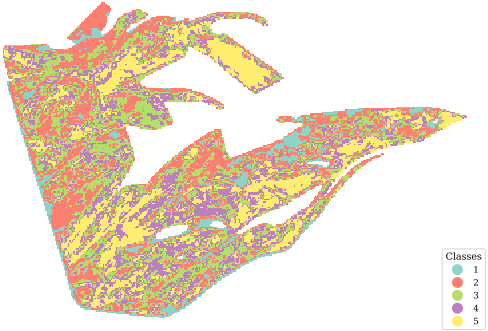
Specific Pipeline Objectives:

1. **Isolation Detection:** Define robust mathematical metrics to extract spatial candidates.
2. **Geostatistical Modeling:** Predict expected physical values using Ordinary Kriging.
3. **Hypothesis Testing:** Formulate a rigorous residual-based testing framework.
4. **Likelihood Reassignment:** Correct physically inconsistent pixels via MLE.

Data Overview: Spatial Distribution



(a) Continuous NDVI Distribution (Z)



(b) Fragmented Categorical Map (L)

- The categorical map (b) exhibits high-frequency noise.

Verification of Prerequisite: Spatial Autocorrelation

Fundamental Assumption

Ordinary Kriging requires the attribute field $Z(\mathbf{X})$ to exhibit spatial autocorrelation, rather than complete spatial randomness [2].

Quantitative Assessment (Global Moran's I):

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$$I = \frac{N}{W} \frac{\sum_i \sum_j w_{ij}(Z_i - \bar{Z})(Z_j - \bar{Z})}{\sum_i (Z_i - \bar{Z})^2} \tag{1}$$

- **Result:** A highly significant positive spatial autocorrelation ($I = 0.8002$, $p < 0.001$).
- **Implication:** This provides the theoretical mandate to rigorously model the physical field $Z(\mathbf{X})$ as a **Spatial Random Field (SRF)**.

Data Normalization

Normality Violation: Original data is strictly non-Gaussian (skewed). This violates the normality assumption in later hypothesis testing, which can lead to biased predictions and invalid hypothesis testing [5].

Normal Score Transformation (NST)

$$Y(\mathbf{X}_i) = \Phi^{-1}(F(Z(\mathbf{X}_i))) \tag{2}$$

* Φ^{-1}: *Inverse standard normal CDF;* F: *Cumulative Distribution Function of* Z.

Transformation Results:

- Regularized spatial skewness to ≈ 0.
- The empirical quantiles align with the Gaussian diagonal (see Q-Q plot).

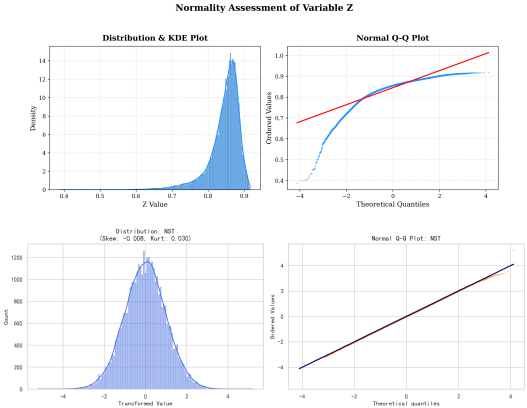


Figure 2: Normality Evaluation: Raw Z (Top) vs. NST Transformed Y (Bottom).

Step 1: Isolation Detection (Candidate Extraction)

Objective: Identify spatially isolated pixels that are likely to be classification errors (artifacts) rather than genuine geographic features (anomalies).

Mathematical Metrics:

- **Neighbor Disagreement Score (NDS):** Quantifies spatial contradiction.

$$NDS(\mathbf{X}_i) = \frac{1}{|\mathcal{N}_i|} \sum_{j \in \mathcal{N}_i} \mathbb{I}\Big(L(\mathbf{X}_j) \neq L(\mathbf{X}_i)\Big) \tag{3}$$

  * *$\mathbb{I}(\cdot)$ is the indicator function; $\mathcal{N}_i$ defines the Moore neighborhood.*

- **Connected Components Analysis (CCA):** Evaluates topological connectivity to flag clusters below the designated Minimum Unit.

Algorithmic Logic

The pixels within the union set $\mathcal{S}_{\text{cand}} = \mathcal{S}_{\text{NDS}} \cup \mathcal{S}_{\text{CCA}}$ are considered isolated and serve as prediction targets in the subsequent Kriging stage.

Step 2: Geostatistical Validation (Ordinary Kriging)

Kriging System

$$\textbf{Estimator: } \hat{Z}(\mathbf{X}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{X}_i) \tag{4}$$

$$\textbf{Variance: } \sigma_K^2(\mathbf{X}_0) = 2\sum_{i=1}^{n} \lambda_i \gamma(\mathbf{X}_i, \mathbf{X}_0) - \sum_{i=1}^{n}\sum_{j=1}^{n} \lambda_i \lambda_j \gamma(\mathbf{X}_i, \mathbf{X}_j) \tag{5}$$

λ_i are the Kriging weights determined by solving the Kriging system of equations, γ is the semivariogram to model spatial dependence.

Why Ordinary Kriging?

- Best Linear Unbiased Predictor.
- Provides physically grounded uncertainty σ_K^2 derived directly from the spatial covariance structure (semivariogram γ).

Computational Optimization: Solving the system of Kriging equations need compute the inverse of the covariance matrix, which is $\mathcal{O}(N^3)$, therefore utilizes only the N nearest neighbors (e.g., $N = 40$).

Step 3: Hypothesis Testing Framework

- H_0: The physical attribute $Z_{\text{obs}}(\mathbf{X}_0)$ is a valid realization generated by the spatial stochastic process of the surrounding class $\mathcal{C}_{\text{surr}}$.
- H_A: The physical attribute $Z_{\text{obs}}(\mathbf{X}_0)$ is drawn from a distinctly different spatial process, making it physically incongruent with its surroundings.

Test Statistic $\eta(\mathbf{X}_0)$:

$$\eta(\mathbf{X}_0) = \frac{Z_{\text{obs}}(\mathbf{X}_0) - \hat{Z}_{\text{surr}}(\mathbf{X}_0)}{\sigma_{\text{surr}}(\mathbf{X}_0)} \sim \mathcal{N}(0,1) \tag{6}$$

** Asymptotic normality is mathematically guaranteed by the prior NST step.*

To avoid the multiple testing problem across all candidates, we apply the Benjamini-Hochberg procedure to control the False Discovery Rate (FDR) at a pre-specified level q (e.g., $q = 0.05$).

Decision Rule

1. **Reject H_0:** Large physical discrepancy $\rightarrow$ **Preserve original label** (Real feature).
2. **Fail to reject H_0:** Small discrepancy $\rightarrow$ **Proceed to Reassignment** (Noise).

Step 4: Likelihood-Based Class Reassignment

Maximum Likelihood Estimation (MLE):

$$\mathcal{L}(C_k) = \frac{1}{\sigma_{C_k}(\mathbf{X}_0)} \exp\left(-\frac{(Z_{\text{obs}}(\mathbf{X}_0) - \hat{Z}_{C_k}(\mathbf{X}_0))^2}{2\sigma^2_{C_k}(\mathbf{X}_0)} \right) \tag{7}$$

- **Structural Regularization:** The leading term $\sigma^{-1}_{C_k}$ mathematically penalizes candidate classes that exhibit high spatial uncertainty or sparse local support.

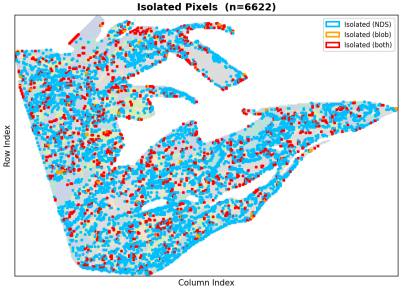
Data-Driven Reassignment

The corrected final label L^* is chosen to maximize the likelihood function across all candidate classes $\mathcal{C}$ (i.e., all 5 classes):

$$L^*(\mathbf{X}_0) = \arg\max_{C_k \in \mathcal{C}} \mathcal{L}(C_k)$$

Result 1: Extraction of Candidate Outliers

(a) Continuous NDS Distribution



(b) Extracted Spatial Candidates

Observations from Step 1:

- **NDS Map (a):** High spatial contradictions are primarily clustered along class boundaries.
- **Candidates (b):** Effectively isolates the "salt-and-pepper" noise patches. **6,622 pixels** flagged as candidates ($\approx 18.4\%$ of total domain).

Result 2: Semivariogram Modeling

Based on the weighted Root Mean Square Error (wRMSE), the Exponential model demonstrated the most optimal fit.

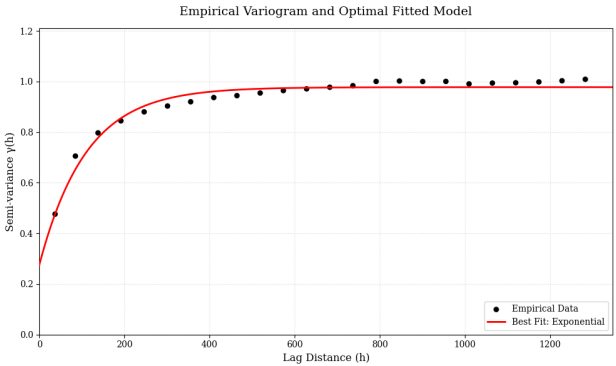
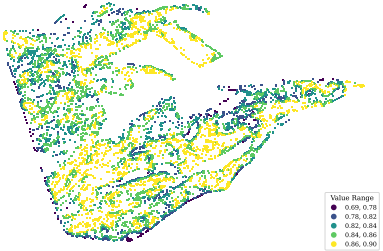


Figure 4: Empirical Semivariogram (black dots) and Fitted Models: Exponential (red).

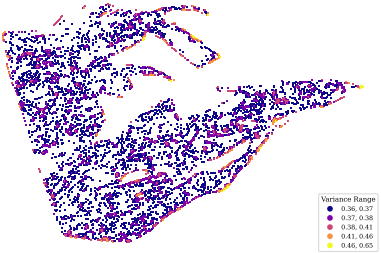
Fitted Parameters:

- **Nugget (c_0):** 0.275, reflects inherent sensor/environmental noise.
- **Sill (c):** 0.978, indicates the total variance of the spatial process.
- **Range (a):** 110.0 meters, defines the spatial correlation distance.

Result 3: Expected Realities and Uncertainty



(a) Kriging Prediction Field ($\hat{Z}$)



(b) Kriging Variance Field (σ_K^2)

- **Prediction (a):** Successfully maps the underlying smooth continuous field behind the categorical noise.
- **Variance (b):** Uncertainty is spatially quantified ($\sigma_K^2 \in [0.36, 0.65]$). High variance zones correspond to areas with sparse data support, guiding the subsequent hypothesis testing.

Kriging provides a physically reliable benchmark for testing.

Result 4: Statistical Decisions (The Filter)

False Discovery Rate (FDR) Control at $q = 0.05$:

Statistical Category	Pixel Count	Percentage
Total Isolated Candidates (S_{cand})	6,622	100%
Rejected H_0 (Genuine Anomalies)	**6**	**0.1%**
Fail to reject H_0 (Artifacts/Noise)	6,616	99.9%

Interpretation & Impact:

- The visual "salt-and-pepper" noise is overwhelmingly confirmed as non-physical artifacts (99.9%).
- Crucially, the conservative FDR test successfully identifies and **preserves 6 legitimate micro-structures** that would have been destroyed by standard filters.

Result 5: Label Correction Analysis (MLE)

MLE Reassignment Summary:

- **Reassigned (5,180):** Transitioned to a new label that maximizes the physical likelihood $\mathcal{L}(C_k)$.
- **Retained (1,436):** Geometrically isolated, yet **statistically most probable** in their original class.

The Defeat of Heuristic Filters

This proves that a pure "Majority Voting" algorithm would have mistakenly flipped the labels of **21.7%** (1,436/6,616) of the isolated pixels.

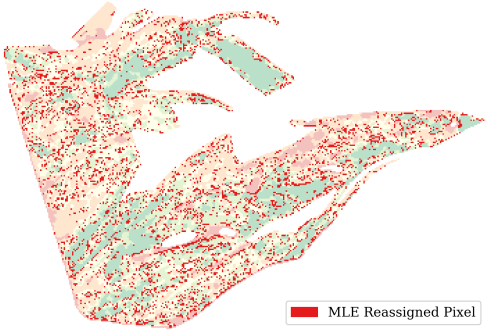
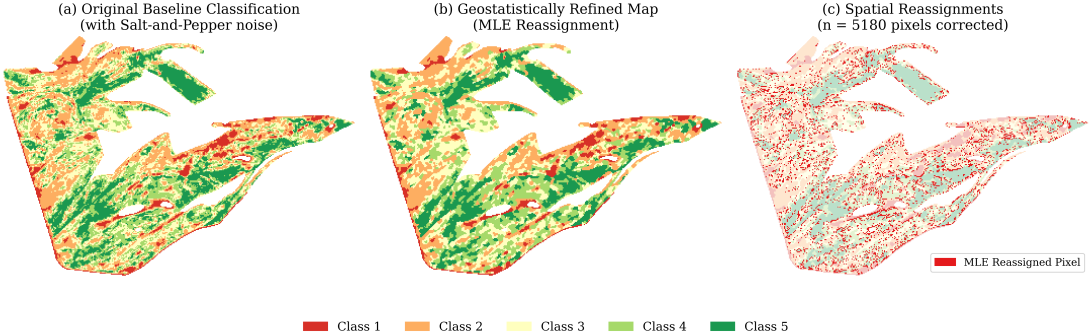


Figure 6: Final Refined Categorical Map.

Comprehensive Map Comparison



(a) Original Fragmented **(b) Refined Map** **(c) Difference Map**

- **Visual Assessment:** The refinement process (b) successfully eradicates spurious noise while strictly preserving crisp linear boundaries and genuine geographic structures.

Discussion and Methodological Limitations

1. **Theoretical and Empirical Discussion**
   - **Validation vs. Smoothing:** The rejection of H_0 for only 0.1% of candidates (Step 3) confirms that 99.9% of isolations were physically consistent with their surroundings.
   - **Retention Rate:** Statistical evidence (MLE) supported the retention of 21.7% (1,436 pixels) of geometrically isolated labels. This confirms that topological isolation is not a sufficient condition for classification error.
   - **Uncertainty Integration:** Using $\sigma_{C_k}^{-1}(\mathbf{X}_0)$ as a precision weight in the MLE framework (Eq. 3.41) objectively penalizes candidate classes with high internal variance or sparse local support.

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2. **Methodological Limitations**
   - **Assumption of Local Stationarity:** The framework relies on a locally constant mean $\mathbb{E}[Z(\mathbf{X})] = \mu$. In landscapes with significant macroscopic gradients (e.g., elevation trends), Kriging residuals will exhibit bias, potentially inflating Type I/II errors.
   - **Computational Complexity:** Unlike $O(n)$ local window filters, the geostatistical approach requires matrix inversion for each candidate class. The $O(m^3)$ complexity (where m is the number of neighbors) limits scalability for massive regional datasets.

Conclusion & Future Directions

Research Summary:

- **Paradigm Shift:** Transitioned post-classification from "geometric cleanup" to **formal spatial inference**.
- **FDR Control and MLE:** Successfully balanced aggressive noise eradication with the preservation of micro-scale geographic realities.

Future Research Directions:

- **Non-Stationary Modeling:** Replace OK with **Universal Kriging** or **Regression Kriging** to incorporate auxiliary variables (e.g., DEM, Soil Moisture).
- **Spatiotemporal Refinement:** Expanding $\gamma(\mathbf{h})$ into **3D semivariograms** $\gamma(\mathbf{h}, t)$ to validate pixels against their historical physical trajectories.
- **Multivariate Co-Kriging:** Utilizing cross-correlation between multiple spectral indices to enhance refinement stability.

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[1] Julian Besag. "On the statistical analysis of dirty pictures". In: *Journal of the Royal Statistical Society: Series B (Methodological)* 48.3 (1986), pp. 259–279.

[2] Noel Cressie. *Statistics for spatial data*. John Wiley & Sons, 2015.

[3] Kwang E Kim. "Adaptive majority filtering for contextual classification of remote sensing data". In: *International Journal of Remote Sensing* 17.5 (1996), pp. 1083–1087.

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[5] Hans Wackernagel. *Multivariate geostatistics: an introduction with applications*. Springer Science & Business Media, 2003.

Thank You!

Questions & Discussion

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